

Chapter-2

Flipkart E-Commerce Platform Analysis

2.0 Introduction

The entities required for the **Flipkart E-Commerce Platform Analysis** are identified based on the business requirements of an online shopping system. Each entity is defined with its attributes, primary keys, foreign keys, and attribute types to ensure accurate data organization and efficient database management. The relationships among the entities are established to represent the interactions between customers, sellers, products, orders, payments, and deliveries. This analysis provides a clear understanding of the system's data requirements and serves as the foundation for designing the Entity Relationship (ER) diagram, relational schema, and database implementation while ensuring data integrity, consistency, and efficient management of e-commerce operations.

2.1 Entity: Customer

Attribute	Key type
Customer_ID	Primary key
Customer_Name	Not Null
Email	Unique key
Phone_number	Unique key
Address	Not Null
password	-

Entity: Seller

Attribute	Key type
Seller_ID	Primary key
Seller_Name	Not Null
Store_name	Unique key
Email	Unique key
Phone_number	-

Entity: category

Attribute	Key type
category_ID	Primary key
Category_Name	Unique key
Description	-

Entity: product

Attribute	Key type
Product_ID	Primary key
Product_name	Not Null
Category_ID	Foreign key
Seller_ID	-
brand	-
price	-
stock	-
Description	-

Entity: Cart

Attribute	Key type
Cart_ID	Primary key
Customer_ID	Foreign key
Product_ID	Foreign key
Quantity	-
Total_price	-

Entity: Order

Attribute	Key type
Order_ID	Primary key
Customer_ID	Foreign key
Order_Date	-
Order_Status	-
Total_Amount	-

Entity:Order_Item

Attribute	Key type
OrderItem_ID	Primary key
Order_ID	Foreign key
Product_ID	Foreign key
Quantity	-
Unit_price	-

Entity:Payment

Attribute	Key type
Payment_ID	Primary key
Order_ID	Foreign key
Payment_Method	-
Payment_Status	-
Payment_Date	-
Amount	-

Entity:Review

Attribute	Key type
Review_ID	Primary key
Customer_ID	Foreign key
Product_ID	Foreign key
Rating	-
Review_Text	-

Entity:Delivery

Attribute	Key type
Delivery_ID	Primary key
Order_ID	Foreign key
Delivery_Address	-
Delivery_Date	-
Delivery_Status	-

2.2 Entity Relationships

Entity 1	Relationship	Entity 2	cardinality	participation
Customer	Places	Order	One-to-Many	Partial/Total
Customer	Owns	Cart	One-to-One	Partial/Total
Customer	Writes	Review	One-to-Many	Partial/Total
Seller	Sells	Product	One-to-Many	Total/Partial
Category	Contains	Product	One-to-Many	Total/Partial
Cart	Contains	Product	Many-to-Many	Partial/Partial
Order	Includes	Order_Item	One-to-Many	Total/Total
Product	Appear In	Order_Item	One-to-Many	Partial/Total
Order	Has	Payment	One-to-One	Total/Total
Order	Has	Delivery	One-to-One	Total/Total
Product	Receives	Review	One-to-Many	Partial/Total